Final Design Report

Team 73 - EMBR Damping

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Acknowledgements

Through the development of this project, the team has received support from various people and entities that need to be acknowledged. These parties have significantly helped the progress of the project, and this project would not be possible without their support.

From the beginning stages of the project, our client supplied the materials and parts the project was to be based upon. This included a complete FOX 36 Rhythm fork, equipped with their GRIP damping system. An additional GRIP damper was also included, with the intention of providing a system to keep stock and a system to modify. A few other components aligned with data acquisition were also included, involving a series of sensors and an acquisition unit created by our client.

When deciding on fluid, ARUS MR TECH was kind enough to offer a generous discount on their products. This allowed the team to spend a moderate amount on the ferrofluid, whereas without this discount, we would have most likely exhausted our entire budget simply on fluid. We want to thank ARUS MR TECH for their support in the development of this project.



Executive Summary

This project aims to design, prototype, and validate an electromagnetic mountain bike suspension damper capable of providing variable damping using magnetorheological (MR) fluid and an actively controlled electromagnetic field. The main objective is to demonstrate that damping force can be modulated electronically in real time, providing performance benefits beyond those achievable with traditional mechanical adjustment.

The development process followed an iterative, experiment-driven approach. The initial V1 prototype focused on ease of integration by retaining much of the stock damper and implementing an external electromagnetic coil to be the least complex. While this design did preserve baseline damping behavior, testing revealed negligible damping change when current was applied to the electromagnet. Analysis and further research indicated two critical shortcomings. The first was insufficient flow restriction of the fluid, and the second was improper magnetic field line orientation relative to fluid flow.

These findings informed a complete redesign for the V2 system that introduced a fundamentally different mechanical and electromagnetic system. The key improvements included an internal electromagnetic coil wrapped around a steel piston to direct magnetic flux, an annular gap between the piston and the inner cartridge designed to increase fluid shear under perpendicular magnetic field lines, and an external steel collar that was used to direct the field lines outward to obtain that perpendicularity. Computational fluid dynamics (CFD) modeling and analytical calculations guided the refinement of piston geometry which had the goal of balancing baseline damping performance with activation potential. Electromagnetic modeling confirmed that the redesigned system could generate field strengths sufficient to induce MR fluid yield behavior.

A compact control system was developed to regulate coil activation using real-time sensor data from dual accelerometers. This system enabled active damping behavior by identifying compression and rebound strokes and turning on or off with a changing amount of current proportional to the accelerometer data. This control system also addressed challenges such as power consumption, heat generation, and cavitation. Cavitation mitigation was found to be essential, which required precise timing of coil deactivation during the rebound stroke to maintain consistent fluid behavior.

Experimental validation of the V2 system demonstrated qualitative evidence of variable damping with noticeable change in damping response corresponding to applied current. The final system integrated mechanical, electromagnetic, and control subsystems into a functional prototype with a proposed cost of $1400, which is competitive to similar systems in industry. Overall, this project validates the potential for MR damping technology for mountain bike suspension, and it establishes a strong foundation for future development.

Introduction

The aim of this project was to explore a new avenue of semi-active damping for bicycle suspension. As opposed to traditional passive damping methods, semi-active systems are capable of collecting data about the terrain and behavior of the rider and use that data to make real-time adjustments to the system to optimize performance. For example, if the system understands that the rider is climbing and requires as much pedaling efficiency as possible, it will increase damping to provide additional support. The same can be said for situations in which softer, more responsive suspension is desired, in which case damping would be reduced.

The "new avenue" which this project explored was semi-active damping via magnetorheological (MR) fluid. The exact science behind this fluid is described in subsequent sections, but the fluid behavior is the primary driver for this damping method. The prototype design capitalizes on this behavior. In essence, the properties of this fluid change in the presence of a magnetic field. When no field is applied, it behaves as any other hydraulic damping fluid would (Newtonian fluid behavior). As soon as a magnetic field of sufficient strength is applied, it changes the yield stress of the fluid, requiring a higher amount of applied pressure before fluid begins to flow (Bingham plastic behavior). For a damper, this increase in yield stress translates to more damping. Damping characteristics are automated through controlling the magnetic field electronically.

This technology has been implemented in the automotive industry in high-end luxury vehicles and sports cars. The technology exists and has been proven to work extremely well, but it has not been seen commercially at the miniaturized level for bicycle suspension.

Process and Methodology

An Introduction to Suspension

In order to design a meaningful product, the team had to first become acquainted with existing bicycle suspension systems. This began with an investigation into the importance of suspension design and to whom and how it is impactful. Across almost all vehicle applications, be it a family minivan, a high-performance race car, a motorcycle, or a bicycle, the primary objectives of the suspension system are to improve comfort and control. Once general performance criteria were outlined, exploration of the design of various suspension systems (with a particular focus on the damper) could be conducted.

Although the primary goal of the project was to implement an MR fluid system to vary damping, it was important to first understand how traditional systems worked. After all, the team never intended to construct a damper completely from scratch; the goal was to modify the original donor system to accommodate an MR system, and an in-depth understanding of the donor system was required.

There are two primary components that make up a suspension system. The first is the spring, which could manifest as an actual steel coil or as a pressurized gas chamber (usually air). Its purpose is to support the combined mass of the vehicle and user. Within this project, the primary mass contribution is made by the rider. The second primary component is the damper, which is used to control the spring's response to suspension movement. In practice, damping is how suspension behavior is fine tuned. Too little damping results in excessive oscillation, making the bike feel unsupportive and too reactive. Too much damping may result in harshness and a phenomenon known as "packing out" where upon fast, subsequent hits don't allow the wheel to rebound quickly enough, and the suspension packs lower and lower into its travel.

Applications of MR Dampers

After becoming familiar with traditional suspension dampers and their governing fluid mechanical properties, the team then began researching how MR dampers are currently being used. The primary application discovered was the automotive industry, as mentioned previously, with higher-end vehicles. Specifically, these dampers are found in luxury vehicles such as the Cadillac Escalade and performance cars such as the Chevrolet Corvette and Ford Mustang (McCausland, 2023). MR dampers are additionally used in vibration control and seismic protection systems (Dassisti, Olabi, & Brunetti, 2022).

Design Narrative

Mountain bike (MTB) suspension systems are complex works of engineering designed to improve comfort and control for the rider. MTB suspension systems are no exception to the general suspension performance criteria discussed above. Improvements in damping control correlate with improvements in traction and rider comfort, while also providing a competitive edge for high-performance athletes. Current MTB suspension solutions require manual, static adjustments by the rider. Several limitations are associated with these current solutions, including manufacturing complexity, performance drift, and the requirement for experienced intervention to make adjustments. A semi-active

system resolves several of these limitations as the need for rider intervention is reduced, offering a new level of inclusivity for the sport.

The team's goal was to miniaturize current MR technologies to prove that semi-active damping with MR fluid is possible for bicycles. To achieve this goal, the team emphasized experimentation-based development. Of course, a significant amount of research was done prior to any prototyping, but instead of taking both semesters to completely flush out a theoretical design, it was important for the team to investigate the boundaries of the technology via small-scale experiments throughout the first semester.

These small-scale experiments involved testing sample MR fluid and passive magnets, deconstruction of a donated fork / damper, and dynamometer testing with the stock fork. The results were crucial for future design choices. For example, it was realized during small-scale testing that it was necessary to use MR fluid that would *not* remain permanently magnetized and therefore get stuck in a higher viscosity state. It was also realized that the stock damper componentry was reasonably complex, and that attempting to control both compression and rebound damping would be beyond the scope of the project. This is consistent with other semi-active damping systems for bicycle suspension, where only the adjustment of compression damping is automated (Fox Float X Live Valve Neo, 2023) (ZEB Ultimate Flight Attendant, 2019).

Having performed initial research and experimentation, it then became important to separate the project into key subsystems. This was extremely important for overall efficiency, and necessitated collaboration because each component of the project was highly interrelated. This meant that if development was made in one area, it was necessary to consider the effect of that development on other subsystems.

Subsystem Overview

MR Fluid

The fluid is what makes this project and this approach to damping unique. MR fluid typically utilizes a silicone or oil carrier fluid and is infused with iron or other ferrous particles. The fluid used here, manufactured by Arus MR Tech, consists of about 76 wt% iron particles, with the remaining 24 wt% consisting of the carrier fluid and other additives to reduce particle agglomeration and particle-carrier separation (ARUS MR TECH).

When a magnetic field is present, the particles arrange in such that they are parallel with the magnetic flux lines. If the field is stronger, the magnetic forces that align the particles also become stronger, resulting in an increase in the fluid's yield stress. When

fluid flows, the molecules that make up the fluid are shearing, and it is this resistance to shear that is associated with the fluid's viscosity. High resistance to shear results in a highly viscous fluid. This property change is represented in the figures below.

MR fluids behave as any Newtonian fluid does in their inactivated state, represented by the "Without Magnetic Field" diagram below. In a damping application, assuming the fluid-restricting geometry remains the same, the force vs. velocity relationship is linear and has a y-intercept of nearly zero. In the activated state, MR fluids behave as a Bingham plastic, or semi-solid. This is similar to Newtonian behavior in the sense that there is still a linear force vs. velocity trend, but the minimum force at which flow is achieved is no longer zero. This means that MR fluid essentially behaves as a solid until the yield stress in the active zone is exceeded, at which point particles begin to shear (flow).

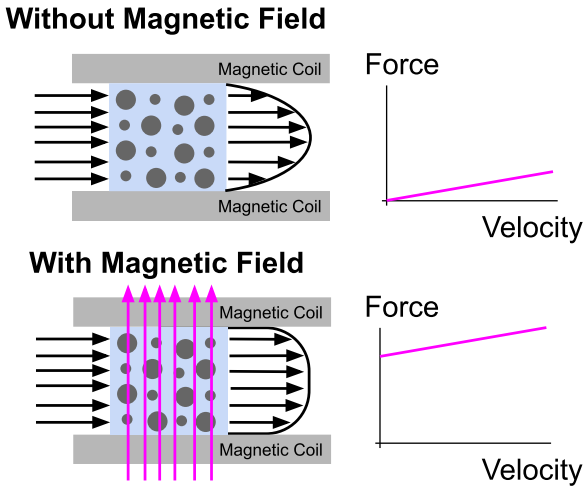


Figure 1: Magnetorheological Fluid Behavior (top) Inactivated (bottom) Activated

More information about the properties of the specific fluid chosen for this final design can be found in the appendix.

Electromagnet

Electromagnetism is the rule behind the physical interaction between electric currents and magnetic fields. According to Ampere's Law, an electrical current flowing through a conductor generates a magnetic field proportional to the amperage. Winding a conductor, such as copper wire, into a coil concentrates this field, and introducing a material with high magnetic permeability, such as steel, directs it. The highly permeable material provides a path of low magnetic reluctance, routing the magnetic flux much like a

wire routes electrical current. This controlled generation and direction of magnetic flux forms the operating principle of our EMBR system.

Mechanical

Although the MR fluid and electromagnet system constitute the primary technology, a significant amount of work went into designing the physical damper geometry to accommodate it.

The stock Fox GRIP damper is made up of three primary systems. The first is the compression damping circuit, which gives the rider targeted adjustment for just the compression stroke. The second is the rebound damping circuit, which allows for adjustment of the rebound stroke. The third and least well-known system is the accumulator, or Internal Floating Piston. Referred to here as an IFP, it serves to account for fluid displacement within the damper cartridge and reduce or eliminate the effects of fluid cavitation. Most modifications made to the damper geometry focused on the compression and IFP systems, largely as a consequence of them being packaged as a coupled unit in the GRIP damper. Reference appendix A1 and A2 for supporting drawings.

At a high level, the mechanical design rationale was centered around realizing the effects of the electromagnet-fluid interaction. In addition to this, the design had to also consider more fundamental properties of fluid mechanics, which quickly became anything but simple.

Control and Power

The control system is responsible for powering and controlling the electromagnet coil in real time based on user input and sensor data. Some existing suspension systems like RockShox Flight Attendant and Fox Live Valve use sensors and microcontrollers to tune the damping characteristics in real time. Where this design diverges from these existing solutions is in the method of varying the damping force. While the existing solutions use servo motors to change the size of the orifice that fluid passes through, this design changes the characteristics of the fluid at the orifice while keeping the orifice size constant. While the clearest reason for using a control system in a magnetorheological damper design is to improve performance, there are several other reasons that are equally important.

The first is battery efficiency. If the electromagnet was left constantly active, it would require a significant amount of energy over the course of a ride. Using a control

system allows the coil to be actuated only when it's needed, saving massive amounts of energy and allowing the battery pack to remain relatively compact and lightweight.

Paired with saving energy and prolonging battery life, only using the coil when necessary reduces the amount of heat generation. Heat is already an issue in purely mechanical suspension systems, and the introduction of what is effectively a resistive heater does not help the situation. Properly controlling the system limits heat input into the damper.

V1 Prototype

The first iteration of the design aimed to re-invent as few components as possible of the stock GRIP damper to ensure its baseline performance remained. The stock GRIP damper utilizes an orifice and shim stack design to restrict fluid flow. For V1 the stock orifice was retained, and all other components were removed.

Due to spatial constraints and ease of manufacturing, the team decided that the best location for the electromagnet coil was on the outside of the cartridge tube. The coil was strategically placed to have the compression piston centered, which was theoretically the location of maximum magnetic flux. The compression piston was also a steel part, which would work in favor of focusing magnetic flux, further centralizing the active region around the orifice.

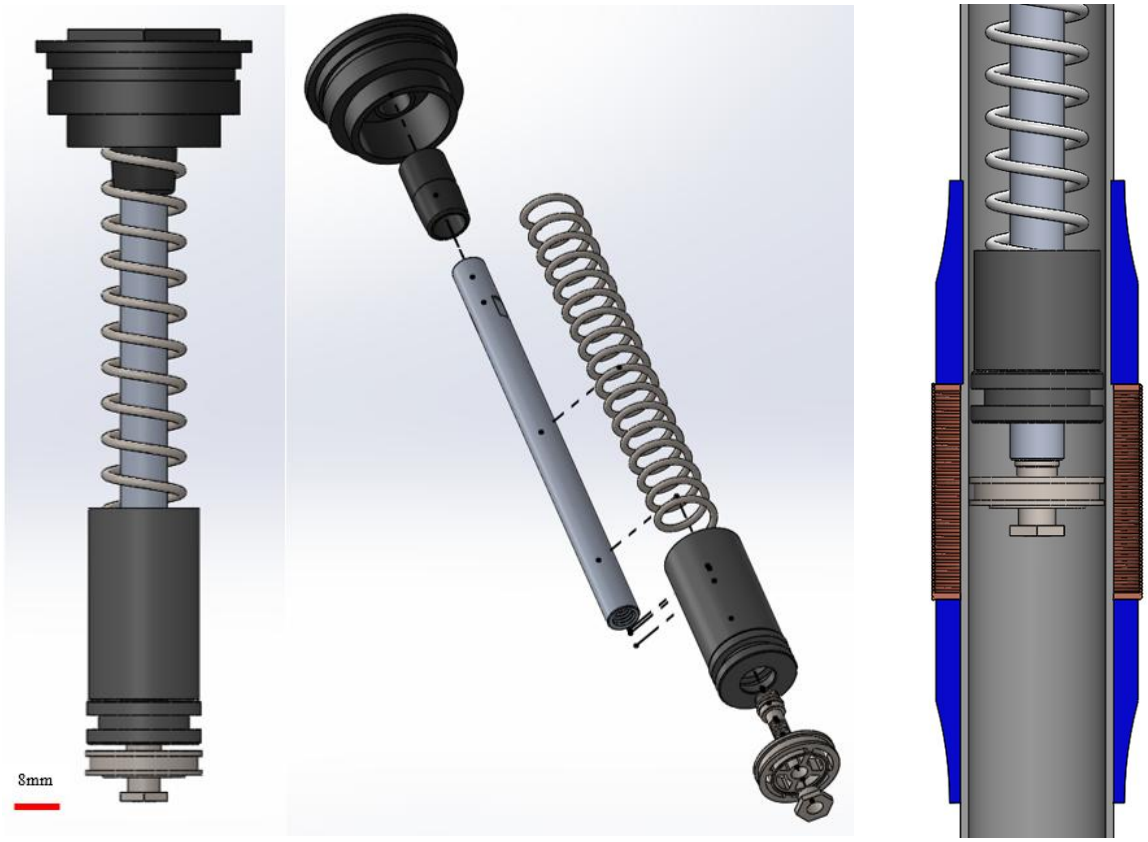


Figure 2: EMBR V1 IFP and Compression Piston Assembly

These decisions were made to be minimally invasive and expedite the testing process, however there were a few key oversights that resulted in an ineffective prototype.

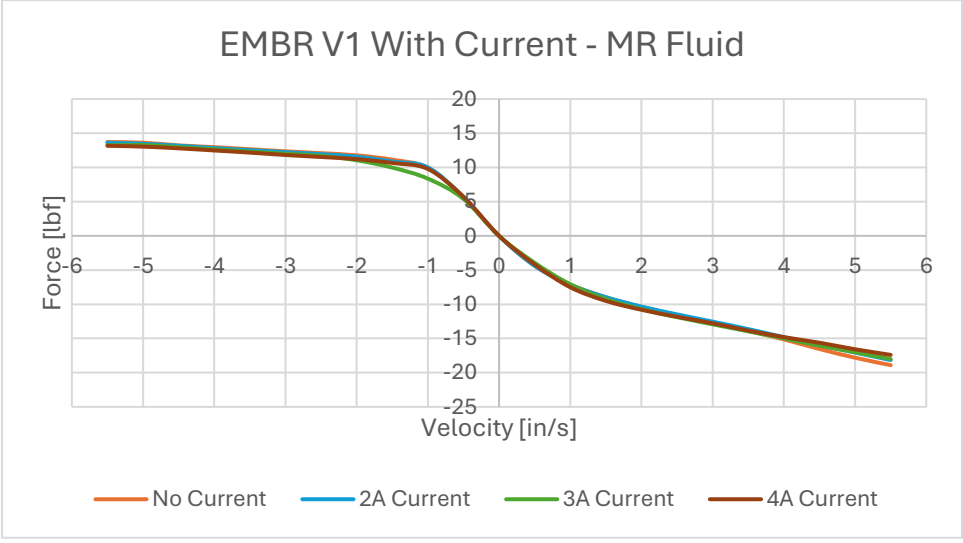


Figure 3: EMBR V1 with MR Fluid with Increasing Current

The results presented in *Figure 3* are a clear indication that even when current was applied, there was little to no effect. The first reason is because the V1 system exhibited extremely low restriction to flow. The second reason has to do with the interaction of the fluid and magnetic field. In the presence of a magnetic field, the iron particles in the fluid align with the field lines. In this case, the magnetic field inside the damper is parallel to fluid flow, as shown below in Figure 4. When the particle chains are parallel with fluid flow, there is little to no change in shear force. Through this, the team learned that it is imperative for the magnetic field to be perpendicular to the fluid flow inside the damper (Dong, 2017).

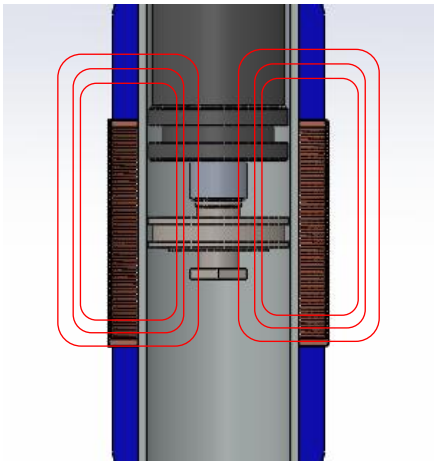


Figure 4: EMBR V1 Magnetic Field Lines

V1 was, for a lack of a better expression, the best attempt at the path of least resistance. From a purely mechanical standpoint, there were design flaws that ultimately resulted in the system not being able to perform. Use of the stock compression orifice was a good starting point in terms of figuring out exactly how damping is generated and why so many commercial MR damping systems are designed the way they are.

V2 Prototype

After learning from the V1 design, the team went on to develop the V2 design with the mechanical and magnetic field goals in mind. The team began by diving into research again to look for current solutions to base the new design from. A Computational Fluid Dynamics analysis, or CFD analysis, was performed to isolate the effects of piston geometry, with fluid calculations being done to validate the simulation results. Multiple pistons were 3D printed and tested to further validate the fluid simulations and

calculations. Finally, a MATLAB model of the electromagnet and its interaction with surrounding materials was created.

V2 Design Improvements:

The first improvement made in the V2 design was to increase compression flow restriction. With the stock orifice, the non-activated case was majorly underdamped. Increasing the restriction of compression flow will result in this case providing a usable amount of damping. This change manifested in the form of a solid piston that created an annular gap between itself and the damper cartridge tube. This was opposed to the compression piston in the stock damper which created a seal between itself and the damper cartridge tube; fluid flowed through holes in the center of the piston.

The second improvement was to move the coil to the inside of the damper and to create a well-defined magnetic circuit. Wrapping the coil around the outside of the pressure tube resulted in low field intensity and negligible activation, even at high power. By locating the coil internally, magnetic field intensity can be greatly increased.

The third, and most crucial, improvement was to orient the magnetic field lines perpendicular to the direction of fluid flow through the compression piston. This will ensure that damping force will change with electromagnet activation. Specifically, the iron chains that are formed with the electromagnetic field will have to shear in order for fluid to flow.

The team chose this approach in search of success with the V2 system as well as improving the efficiency of all subsystems. This process also increased the teams understanding of MR fluid behavior and current MR designs found in literature. This allowed for a rapid design, manufacturing, and intermediate testing process which ultimately led to a functional prototype.

Computational Fluid Dynamics Analysis

The intent of this CFD study was to help guide the development of the V2 damper geometry. Although it may be possible to do in Fluent, it was not of critical importance to model the relationship between the electromagnet and MR fluid. As such, only the fluid behavior in the inactive state was modeled. Considering a damping fluid of identical Newtonian properties to the selected MR fluid, the team was unsure how to design a piston head that would be slightly underdamped in the inactive state while maintaining a small enough annular gap to ensure significant activation could still occur. If the piston diameter

was too large and restricted flow too much when the electromagnet was off, the fork would feel harsh and uncomfortable. If the piston was too small, the field strength would drop off dramatically, and sufficient shearing of the fluid would not occur. By modeling a series of geometries in Fluent, a better understanding of general behavior was achieved.

A simple geometry was constructed within the Fluent Design Modeler, consisting of a section of the cartridge tube, the 8mm IFP shaft, and 20mm tall piston head (diameter varies). All void space was then defined as a fluid body, shown in green in Figure 5.

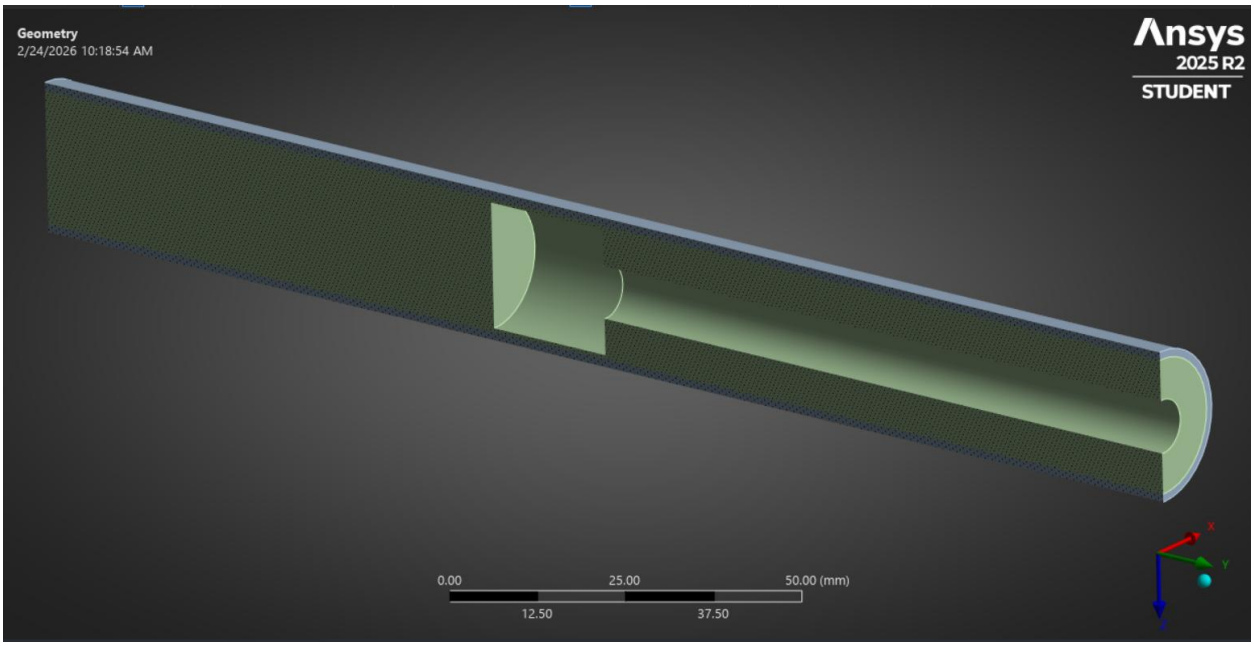


Figure 5: CFD Base Geometry

The piston head and IFP shaft bodies were initially modeled separately from the cartridge tube. A fill operation created the fluid body (green), which then allowed for a Boolean subtraction of the piston head and IFP shaft from the fluid body. This is why these parts appear as empty space. For the purposes of this simulation, that is perfectly fine, as they function as solid boundaries.

Although this geometry is oriented with the Y-axis horizontal, in practice it would be vertical. Referring to model as oriented here, compression flow is left to right, and rebound flow is right to left.

Once the geometry was created, it could then be meshed. A full mesh convergence study was not performed, but it was not necessary given the accuracy of results when compared to hand calculations. A simple mesh refinement was placed around the piston head, as this is the most critical region for flow analysis. Steep gradients were expected

with the sharp flow contraction and expansion on either side of the piston, so a refined mesh was of utmost importance.

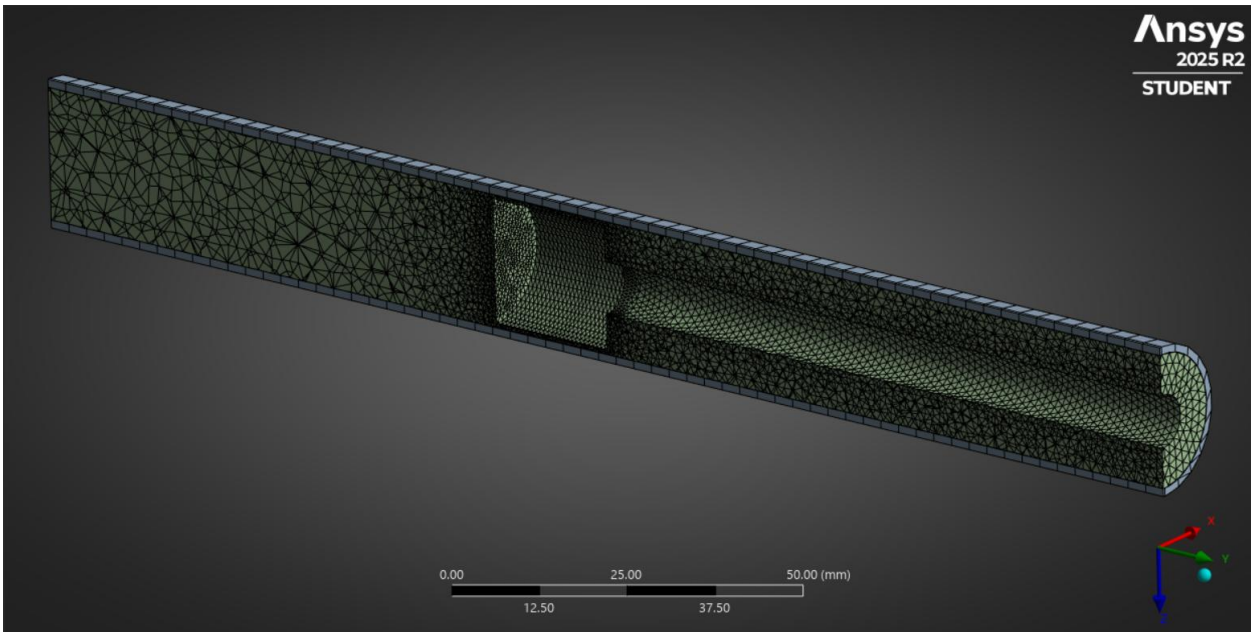


Figure 6: Base Geometry Mesh

Once meshed, the geometry could then be brought into the solver, and the simulation parameters could be defined. A pressure-based solver was used with an SST k-omega viscous model to account for potential turbulent flow. No heat transfer analysis was conducted in this initial study, so the energy equation was left off. Fluid properties were changed to match the MR fluid data sheet, constituting a density $\rho = 2650$ kg/m^3 and a dynamic viscosity of $\mu = 0.055$ Pa-s. The inlet was defined as a velocity inlet, where a range of values were used from 0.5 – 2 m/s. This velocity does not correlate with the fork velocity at the axle, so when comparing the simulation to actual fork performance, a flow rate conversion was required. It was found that a ratio of 0.245 could be used to relate fluid velocity through the cartridge tube to the velocity of the axle.

The most important result of this simulation was the pressure drop across the height of the piston. This is the main damping driver, with a greater pressure differential constituting a greater damping force. The cut plot below demonstrates this pressure drop when fluid is flowing in the positive Y-direction as it would on the compression stroke. This visual represents an 18mm piston, corresponding to an annular gap of 1.1mm.

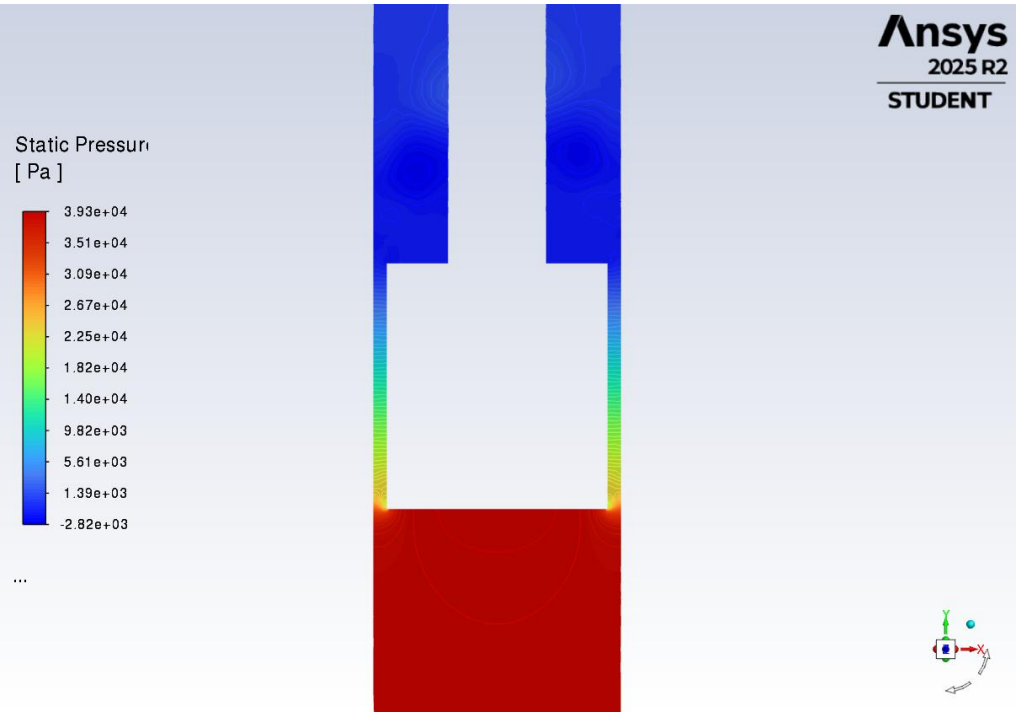


Figure 7: Static Pressure Plot 18mm Piston

Several piston diameters were tested, including 19mm, 18.5mm, and 18mm. This range was largely informed by a second mode of fluid mechanics analysis. This was performed by hand and coded in MATLAB. Because the ratio of piston diameter to cartridge tube ID is close to 1, the key assumption was made that the annular gap could be modeled as a simple rectangular channel. Using the Plane Poisseuille flow model, a MATLAB function was created to report damping force vs. velocity (Deville, 2022). This function was then applied to multiple piston diameters. It became clear that as piston diameter approached cartridge tube ID, the resultant damping force increased exponentially.

This behavior showed that for smaller piston diameters (<18.5mm), even at extreme velocities there was little increase in damping. Increasing diameter by 0.5mm from 18.5mm to 19.0mm, however, resulted in a significant increase in damping force, and from 19.0mm to 19.5mm resulted in an even larger increase. This is largely because damping is calculated by taking the product of pressure drop across the height of the piston and area of the piston face, meaning that a linear change in piston diameter results in a squared change in damping.

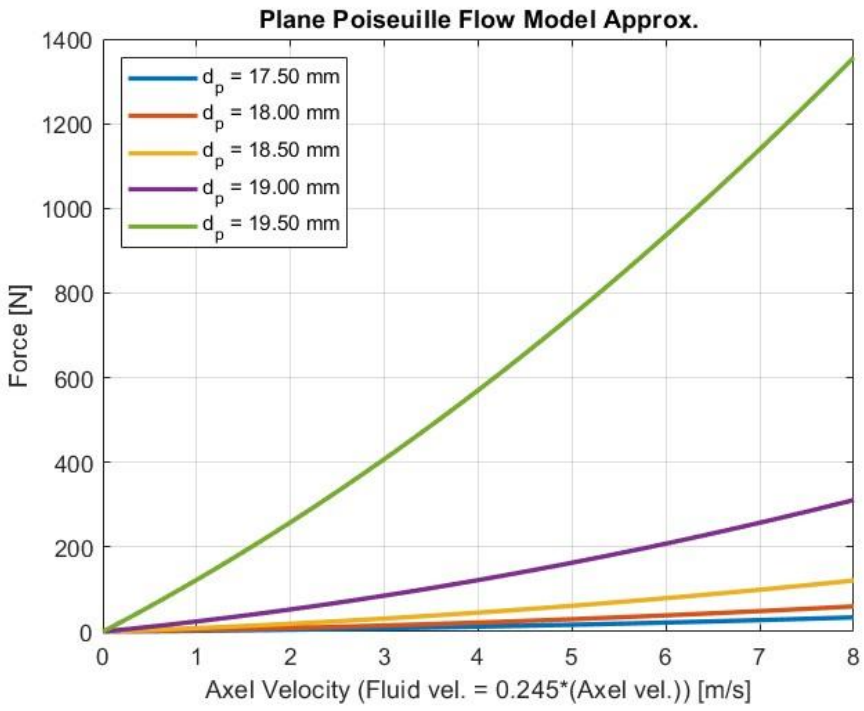


Figure 8: Velocity vs. Force plot

The slight deviation from a linear distribution is the result of minor losses. In this system, minor losses are due to the sudden contraction and sudden expansion experienced as fluid enters and exits the annulus. These losses effectively result in more damping due to turbulent flow, and they become more pronounced at higher fluid velocities.

A piston diameter of 18.5mm was determined to provide the least amount of damping (for the inactivated case) while maintaining a sufficiently small annular gap. This became the primary geometry modeled in Fluent, which could then be compared to and verified by the Plane Poisseuille flow model. The results of this CFD study can be found in the table below.

Table 1: Theoretical Damping Force Results and Model Agreement (d = 18.5mm)

Axle Velocity [m/s]	Plane Poiseuille Flow Model [N]	Ansys Fluent CFD [N]	Model Difference
1	8.42	6.71	20.3%
2	18.75	15.65	16.5 %
4	45.15	40.21	10.9 %
6	79.18	73.48	7.21 %

A series of qualitative tests were also performed to validate these results. These consisted of 3D printing a series of mock pistons in PLA and installing them in the damper. The damper was then cycled by hand. The non-linear force trend demonstrated above was clearly evident in these tests.

Final Mechanical Design

With the geometry verified via CFD simulations and theoretical hand calculations, new components could then be finalized in CAD and manufactured. The V2 design was relatively simple. The most crucial part was a new 18.5mm diameter steel piston that served as the electromagnet core. A new aluminum IFP shaft was also machined. The electromagnet wire was evacuated from the damper via a precisely placed hole in the piston that leads into the hollow IFP shaft.



Figure 9: Final V2 Internal System

Not pictured above is the steel sleeve that was precisely manufactured to slide over the outside of the cartridge tube. This served to complete the magnetic circuit required to produce an adequately strong magnetic field. This is discussed further in the following section.

Final Electromagnet Design

In parallel with the development of the damper geometry, a new electromagnet system was also devised. These two components were highly interconnected, and when fully integrated they both needed to accommodate fluid mechanics and allow for sufficient MR fluid activation.

The hardware consists of a 160-turn, 26-AWG copper coil wrapped around a 9 mm steel core. When energized, this core acts as a concentrator for magnetic flux directing the generated magnetic field. This system is housed within a 19 mm inner-diameter, non-magnetic aluminum tube, allowing the magnetic flux to interact with the enclosed ferrofluid. This orientation of the electromagnet and steel core creates a magnetic field

perpendicular to the fluid flow at either end of the annular gap. Finally, an outer steel shell completes the circuit by providing a low reluctance return path, containing the flux within the active region to maximize the overall electromagnetic effect.

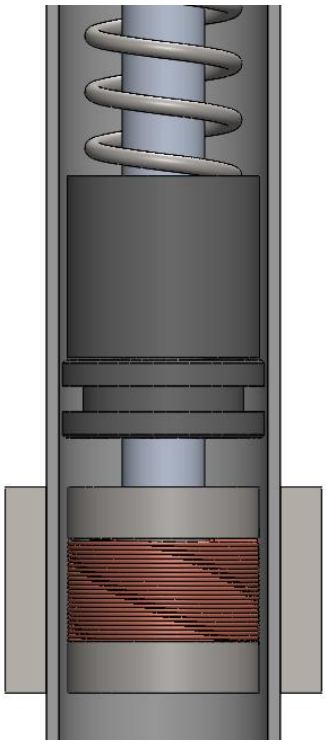


Figure 10: Complete Internal and External System Including the Magnetic Circuit

Powered by an 8V DC supply drawing approximately 2A of current, the system's magnetic circuit was evaluated using a 2D nonlinear Finite Element Method (FEM) analysis. MATLAB's partial differential equations (PDE) toolbox was used for this. The model calculates the distribution of magnetic flux by solving Maxwell's equations across various regions of varying magnetic permeabilities. The figure below shows the magnetic flux path generated by this model.

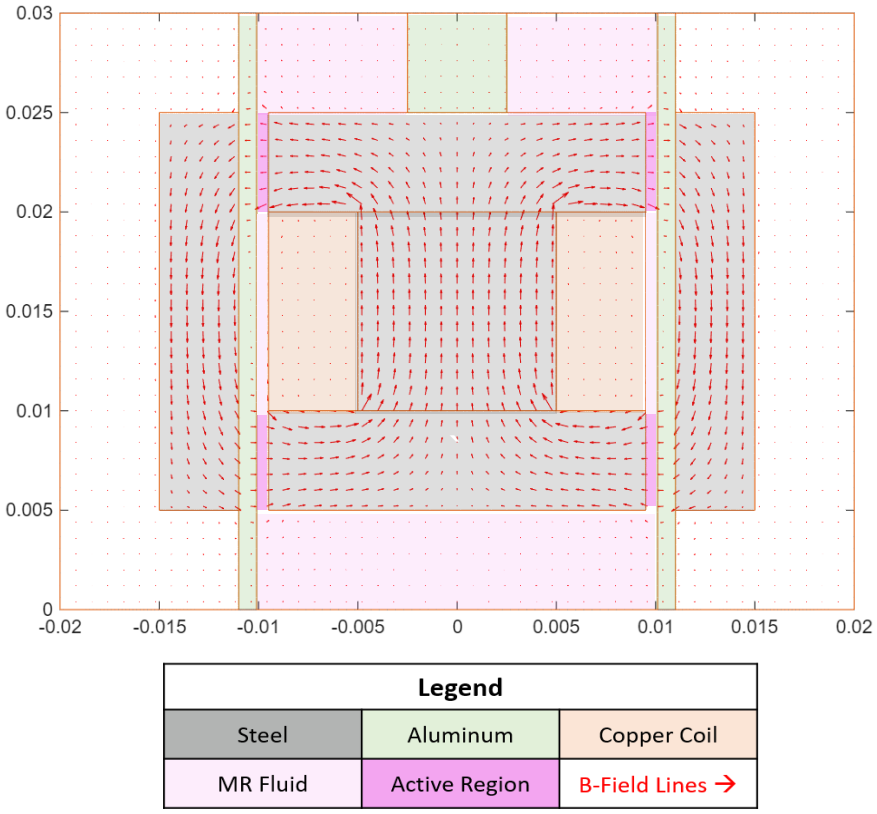


Figure 11: MATLAB Model of Magnetic Flux

Final Control System

In addition to the reasons listed previously regarding the necessity of a control system, the last and arguably most critical reason for implementing a control system is related to cavitation. This was discovered during V2 development and became one of the most important considerations for creating a functional system. In a traditional mechanical damper, the oil that is forced past the compression piston and into the IFP is allowed an alternate and less restrictive exit path. This allows the fluid in the IFP to quickly return to the lower side of the compression piston when the fork rebounds. Cavitation occurs when fluid does not exit the IFP fast enough and a vacuum forms below the compression piston. This phenomenon is shown below in Figure 12. Cavitation becomes a serious issue when the suspension system undergoes multiple repeated compressions. While the first compression may perform as expected, any subsequent compressions will receive little to no damping as much of the fluid is now stuck in the IFP. Because the compression piston in our EMBR damper design does not possess an alternate path for fluid to escape in rebound, the coil must instead be deactivated to avoid cavitation.

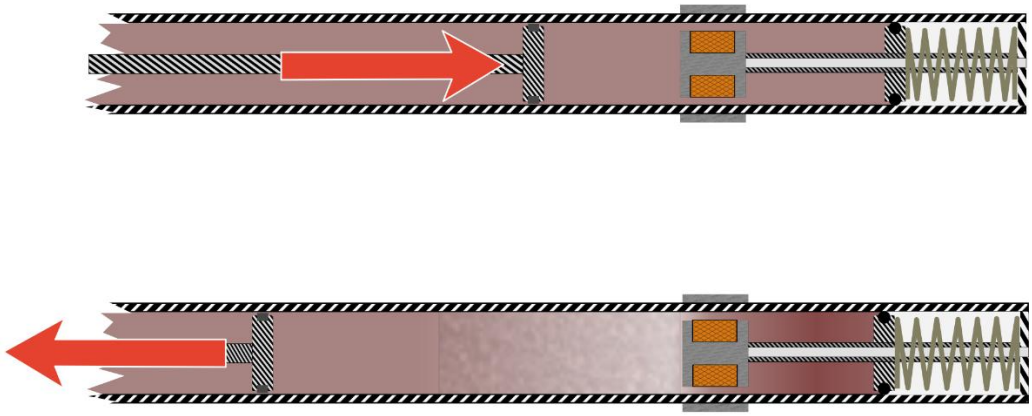


Figure 12: Cavitation in the EMBR Damper

In order to activate the coil when it's needed and deactivate it before it causes cavitation, the control system must be capable of discerning when a compression stroke or rebound stroke begins. Earlier on-bike testing with the data capture tool showed that it was possible to identify these key events using a pair of accelerometers. The current control system considers both the relative acceleration and the derivative of the relative acceleration to identify compression and rebound motion. Shown below in Figure 13 is a 300ms snapshot of accelerometer data with certain regions colored in to indicate where the system would attempt to activate or deactivate the coil.

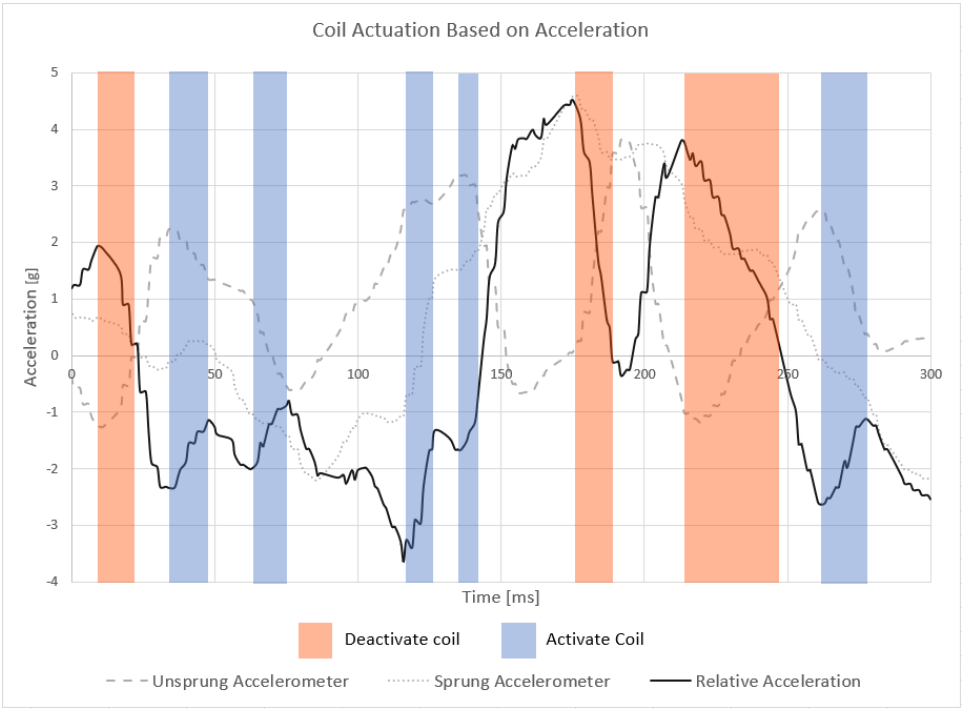


Figure 13: Plot of coil actuation based on relative acceleration

Shown below is the final control system mounted to the prototype fork. It was designed to occupy as little space as possible and not interfere with the normal motion of the fork and bike. The system is equipped with two buttons on top for adjustment, a power switch on the side, and USB-C charging for the pair of 18650 Li-ion cells.



Figure 14: The final control system mounted to the EMBR equipped suspension fork

The Fully Integrated System

The fully integrated system CAD and actual prototype installed on team member Eli Webster's bike can be found below.



Figure 15: Fully Integrated System

Results and Discussion

The resulting success of the final system behavior serves as evidence that an onboard electronic system with the capacity for active damping adjustment using MR technology is a viable solution. These results include the qualitative and quantitative findings that demonstrate the differences in system behavior due to changes in fork accelerations. The first qualitative test found noticeable levels of damping associated with each level of current that the control system was delivering to the magnet. This test also discovered an unexpected behavior of the fork: a pitched frequency, the volume of which was responsive to the amount of current supplied. After some investigating, this frequency was found to be the Pulse Width Modulation frequency that was being used in the control system.

The final system was tested on the shock dynamometer, or dyno, several times. In short, these results proved to be inconclusive and highlighted limitations of the testing setup. The final results are shown below in Figure 16.

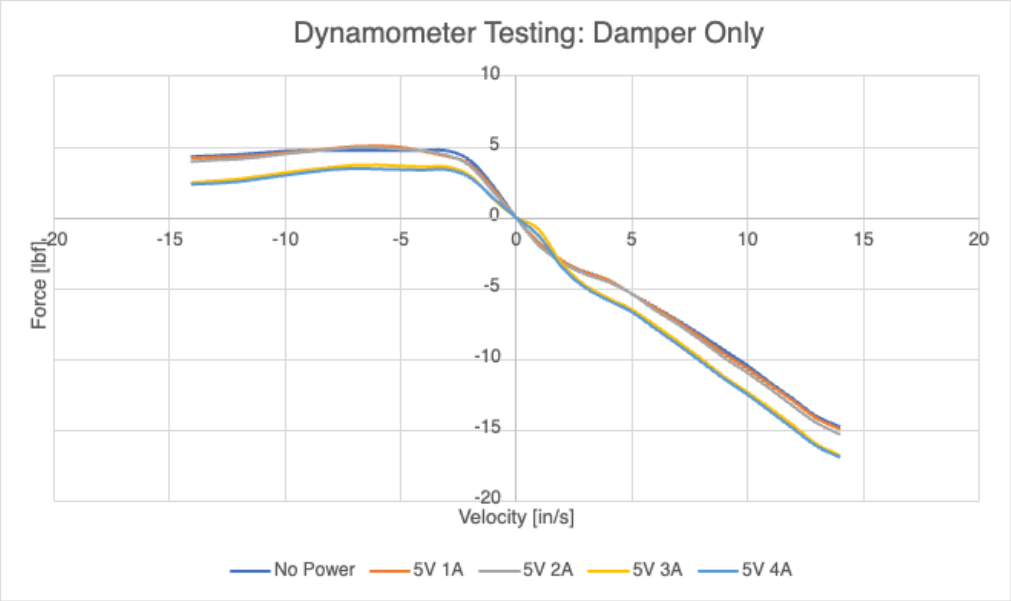


Figure 16: Dynamometer Velocity vs. Force plot

The main limitation of this testing is the maximum velocity of the shock dyno. Mountain bike dampers are typically tested to shaft velocities of 2 m/s, or 80 in/s (Schoeneck, Sadauckas, & Nagurka, 2023). In contrast, the maximum velocity the school's shock dyno can achieve is 14 in/s. As such, the complete picture of the damper's performance remains unknown; only the low-speed damping can be quantified.

The damper was tested in two configurations; with a power supply providing constant power, and with the control system wired up. With the power supply, the damper showed signs of cavitation. This was evident in the fact that compression damping forces

decreased with higher power levels, which is opposite of the expected behavior. This was verified through a visual inspection of the IFP through a port on the side of the damper cartridge tube in a separate test. On the rebound stroke, the IFP spring would not move. This meant that no fluid could flow past the compression piston. If no fluid replaced the volume of the damper shaft as it extended, a vacuum would form causing cavitation. The control system, with an algorithm to identify compression and rebound strokes and activate the damper accordingly, is still in an early phase of development. As such, it was unable to identify compression and rebound on the shock dyno. This can be fixed with further development of the control algorithm. However, this meant that the team was unable to draw any definitive conclusions from dyno testing.

Cost Analysis

The total cost of team-purchased supplies is $415.59. Below in Table 2, an itemized list of purchases is provided. In addition to these parts, our client provided a Fox 36 Rhythm 170mm Fork complete with a Fox GRIP Damper cartridge. These two client-provided parts totaled $560, for a combined total of $975.59.

Table 2: Itemized budget

Item	**Cost ($)**
Damper Clamp	$31.97
Fork Axle	$7.59
MR Fluid	$275.00
Magnet Wire	$12.14
Electrical Components for Control System	$46.74
Metal Rod for IFP Shaft	$9.99
Metal Stock for Damper	$32.16
Team Total:	$415.59
Client: Fox 36 Rhythm Fork 170mm	$400
Client: Fox GRIP Damper	$160
Client Total:	$560
Combined Total:	$975.59

The estimated cost of the complete EMBR fork is $1400. The main factor in this price estimate, besides the cost of the fork chassis and air spring, is the MR fluid. The damper takes 80-100 mL of MR fluid to fill. With this fluid coming in at about $1000 per liter at retail price, this equates to roughly $100 of fluid per fork. Production deals could be made to

bring this price down, but for the sake of getting a current estimate, retail pricing will be used. The electrical components used for the control system come in around $50. Metal stock for the damper as well as the magnet wire will be $20-30. All in all, there will be around $200-250 worth of components added to a stock fork. A retail price of $700-900 for current high-level forks plus a markup for R&D and labor costs brings us to the $1400 estimate for the EMBR fork. A price comparison of electronically-actuated forks is shown below in Table 3.

Table 3: Price comparison for industry competitors

Competition Comparison		
Model	**Price**	**Weight (g)**
EMBR Fork	~$1,400	2360
Rockshox Lyrik Flight Attendant	$1,700	2340
Fox 36 Live Valve	~$1,600	2140

Lessons Learned and Future Work

Throughout the project, there were various lessons learned that would improve performance. One of the major lessons was discovered after the first iteration of the design, when the team learned that it did not work. The major revelation was that the design and behavior of existing MR dampers should have been researched before working on the first design. The team strayed from current designs due to the complexity and size of the parts we were working with, and in turn, the team lost sight of the required aspects of the design which would directly affect its functionality. Due to this oversight, the team then had to dive into research midway throughout the project to pivot the design that more closely resembled current solutions for seismic applications. One of the lessons learned from this was the fact that the failure was found quickly. Instead of spending months to get a working prototype, there was only about two weeks between design finalization and testing. This allowed plenty of time to be spent on research and carefully designing the V2 damper in order to ensure that it would work. The second time around, more effort was spent designing, manufacturing, and testing which contributed to its success.

After completion of the project, there are still plenty of items that could be addressed if the team had another semester. The first one that comes to mind is the inner damper piston geometry. While the least-damped case of the V2 prototype was significantly better than the V1 prototype, it is still not quite optimal. Given more time, several metal pistons of various diameters and designs could be tested with the dyno.

From there, the design could be optimized to provide the same damping levels as the stock GRIP damper's least-damped case.

The second step moving forward is to test the durability and longevity of the product. This could be done by analyzing the fluid under a compound optical microscope and comparing it to the unused fluid. Excess particles in the used fluid would show signs of wear and abrasion of the damper. A further test could be performed to quantify the relationship between damper cycles and wear. This could be used to calculate the theoretical lifespan of the damper.

Additionally, the team could aim for an optimal fluid composition. Specifically, the team is aiming for the best compromise between weight and fluid properties. More background research of fluids could be performed. Given enough time and budget, physical tests could be performed with a variety of fluids to determine the best one for this application. The team did not have the amount of time or the budget for this during this cycle. Additionally, this falls outside the main scope of this project. The team wanted to focus on damper design, with fluid composition and properties taking a backseat.

The final aspect that could be improved in future work is the control system. Significant improvements can be made in the way of higher accuracy/faster accelerometers, a microcontroller than can sample data at a faster rate, the integration of an optical-based position sensor, applying digital signal processing and filtering to the raw sensor data, and a PID based feedback control system. The control system proved the ability to modulate damper behavior and respond to sensor input, but there is a tremendous amount of potential future work to fully test and tune the system to compete with current Rockshox and Fox products.

Works Cited

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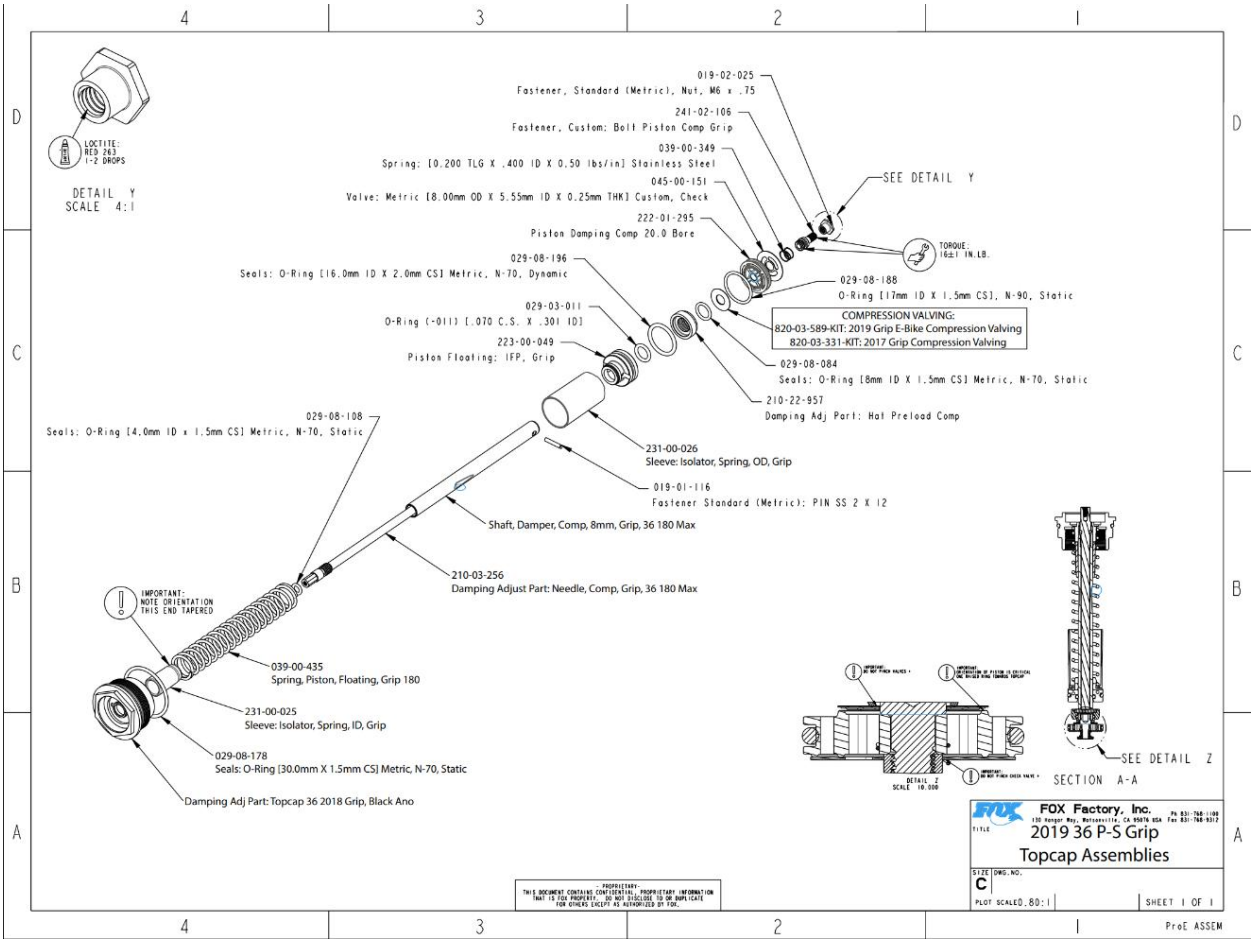
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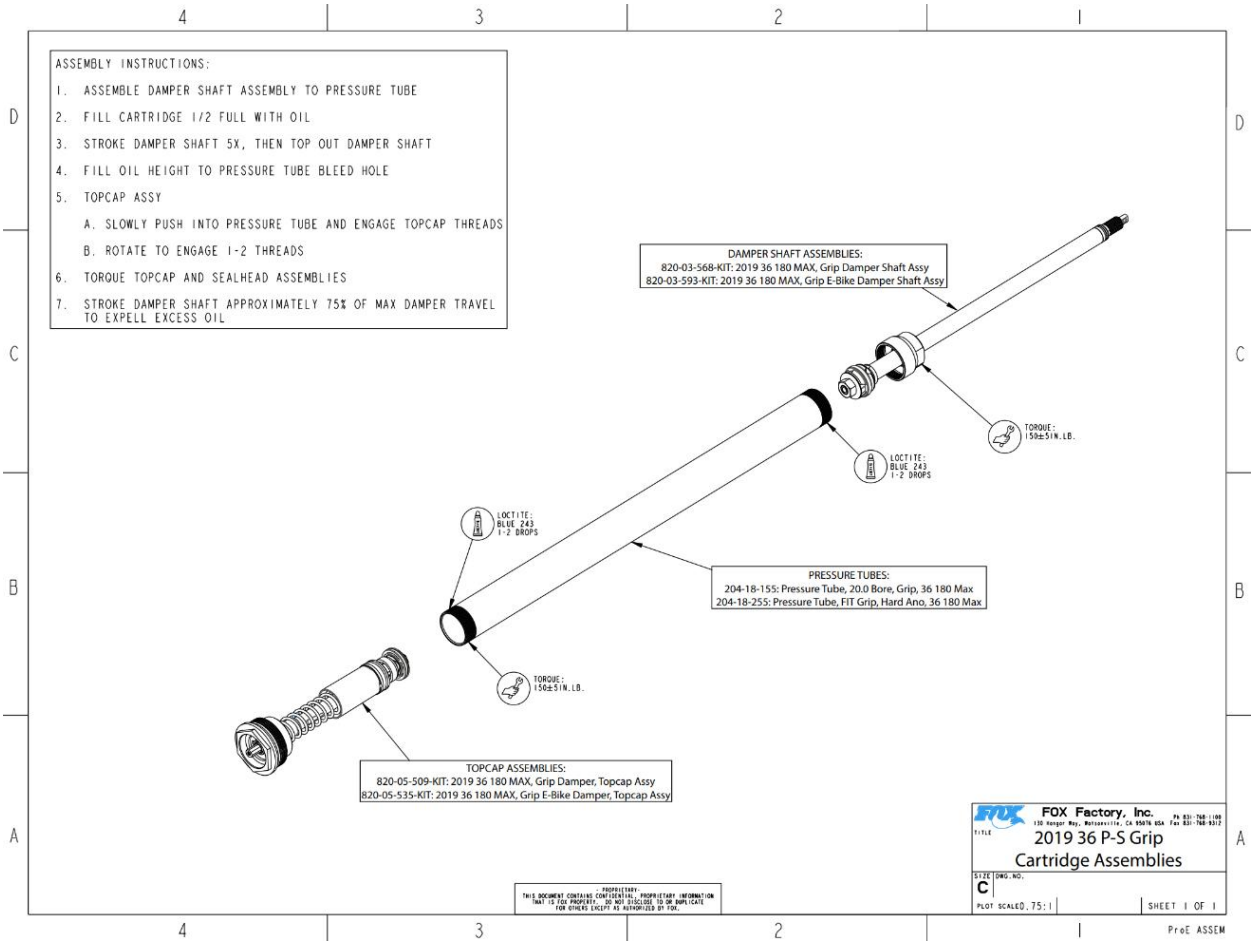
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Appendix



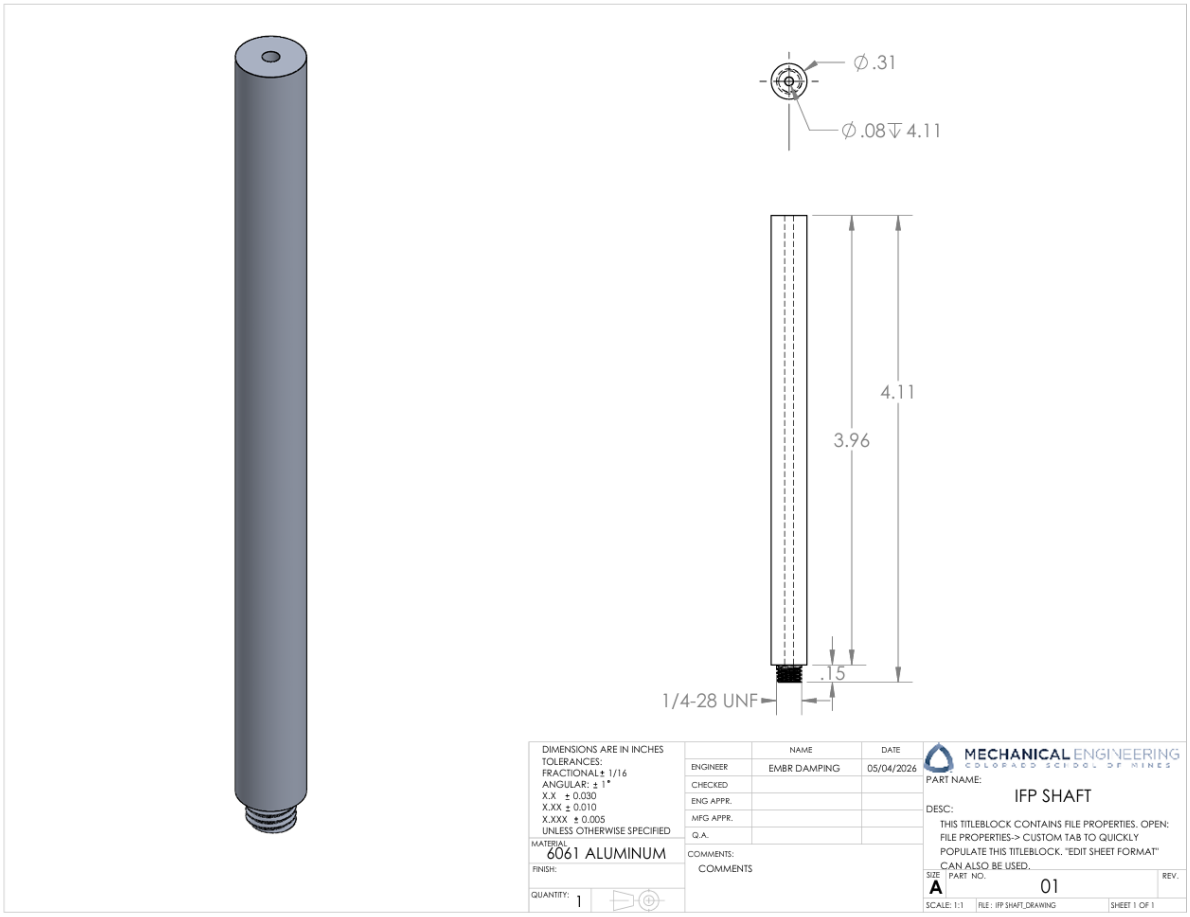
A 1: 2019 36 P-S Grip Topcap Assemblies



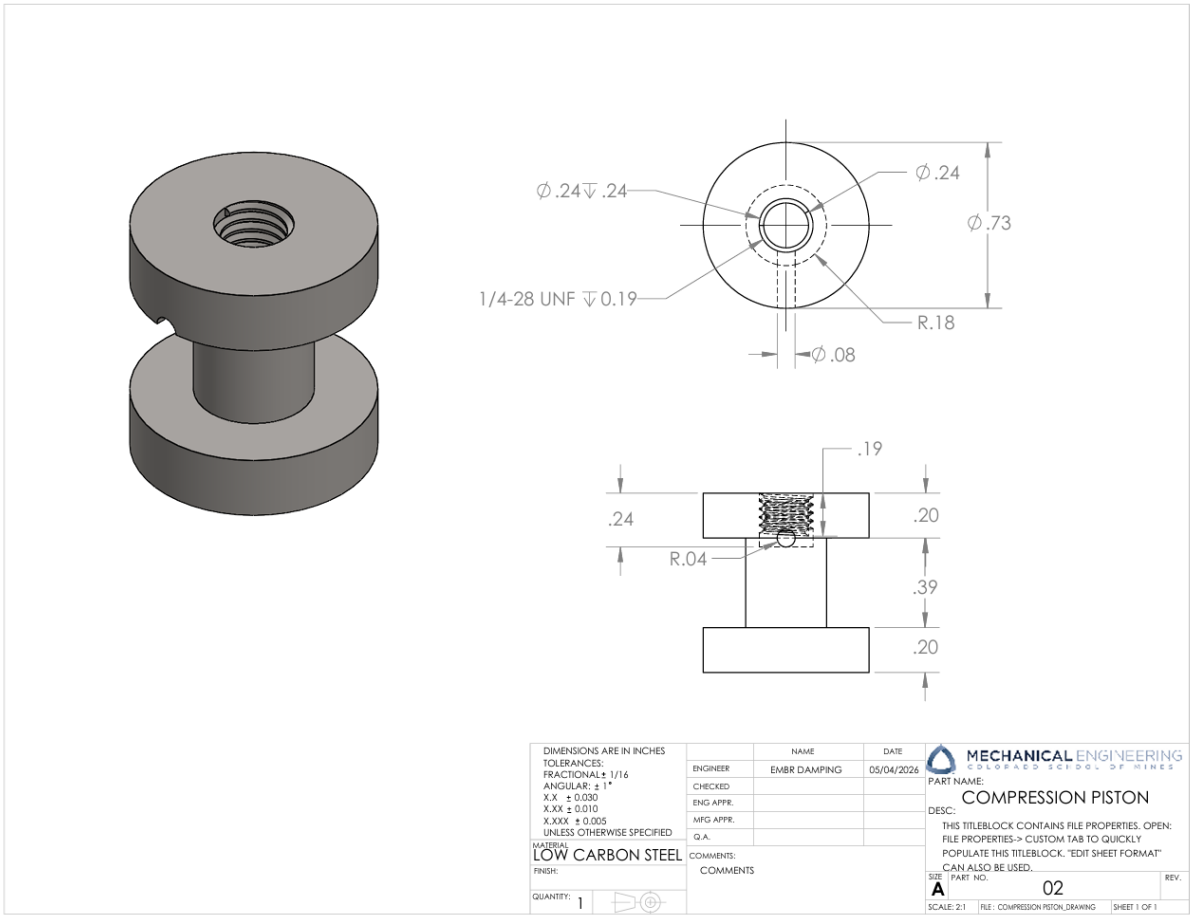
A 2: 2019 36 P-S Grip Cartridge Assemblies

Appearance	Dark Gray Liquid
Density (g/cm³)	2.50 - 2.65
Viscosity (Pa.s) 40°C (500 - 800 1/s)	0.055 ± 0.015
Response Time	< Milliseconds
Flash Point(°C)	>180
Operating Temp Range(°C)	- 40 to +150
Solid Content by Weight %	76
Max.Yield Stress (kPa) @ 100 1/s, 180 kA/m	47 ± 5

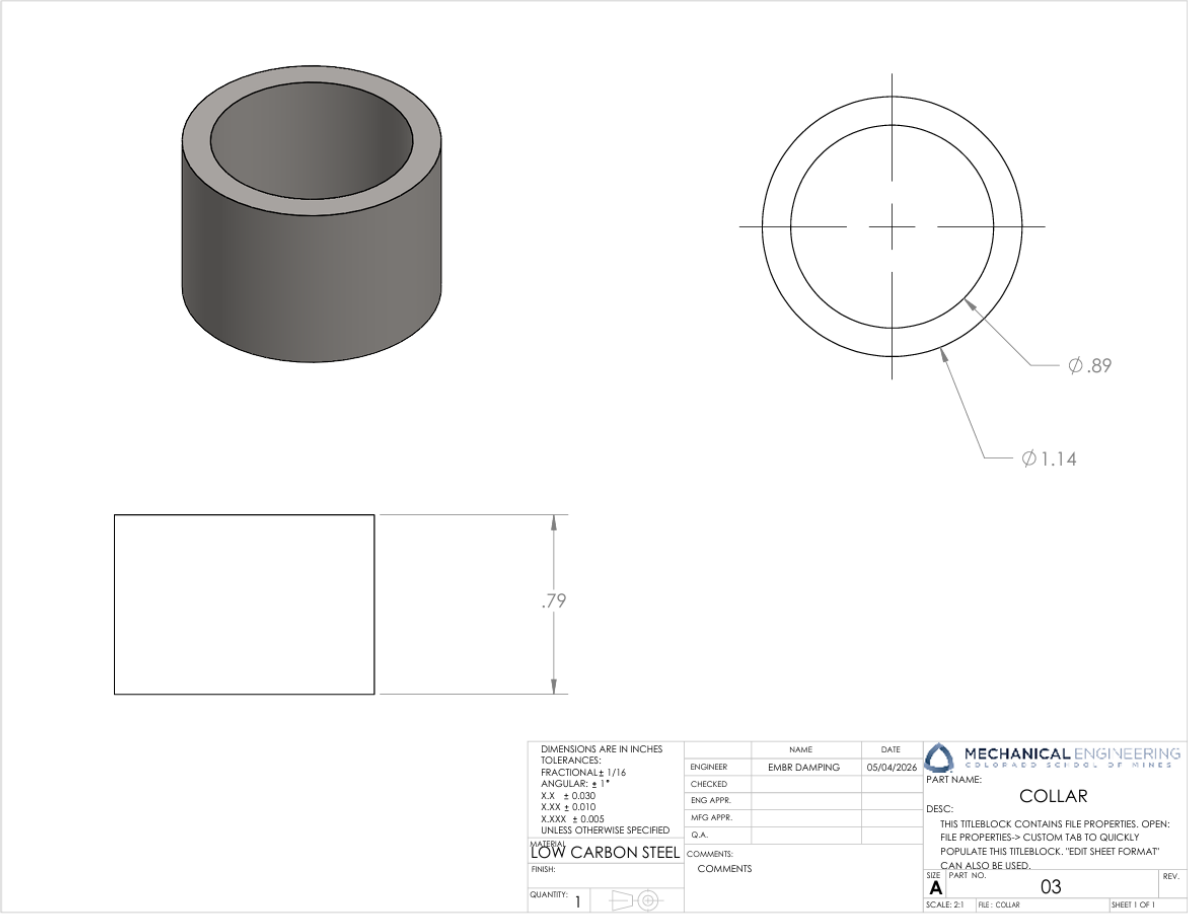
A 3: Fluid Properties for Chosen Fluid



A 4: IFP Shaft Final Engineering Drawing



A 5: Compression Piston Final Engineering Drawing



A 6: Collar Final Engineering Drawing

```
//Graham Adleman - May, 2026
//EDNS492 - EMBR damping - Controller Code V1.4 - Showcase edition
//accel
#include "FastIMU.h"
#include <Wire.h>
#define IMU_ADDRESS 0x69
// create accelerometer 1 and 2 on the two available i2c busses
BMI160 IMU1(Wire1);
BMI160 IMU0(Wire);

#include <SPI.h>
#include <SD.h>
const int chipSelect = 10;

// +++++++++ TUNING +++++++++
const float ACCEL_SEN =0.1; // minimum relative acceleration to activate the coil
const float D_ACCEL_SEN = 0; // minimum relative jerk to activate coil




// main button to start/stop data capture
const int button1=2; // pin for left hand button on physical system. toggles data logging.
const int button2=3; // pin for right hand button on physical system.
const int magPin=5; // pin for pwm signal to magnet coil MOSFET
int magPWM = 0;

// data capture variables
long time0; // ms at time of data capture
long time1; // ms at time of previous data capture
long timeCycle = 0; // timer used in cycle showcase mode 3
float accel1; // unsprung accleration
float accel0; // sprung acceleration
float relAccel = 0; // relative acceleration. calculated using accel0 and accel1.
float lastRelAccel = 0; // previous relative acceleration. used to calculate rate of change.
float dRelAccel = 0;
long timeOn = 0;

bool dataCapture = 0; // toggles data logging on and off

int coilMode = 1; // integer value of the current mode. toggled with button2.
int nModes = 3; // integer number of total coil modes.

File dataFile;
```

```
// setup for IMU
calData calib = { 0 };  //Calibration data
AccelData accelData;    //Sensor data
GyroData gyroData;
MagData magData;

void setup() {
  // physical buttons
  pinMode(button1, INPUT_PULLUP);
  pinMode(button2, INPUT_PULLUP);

  // magnet PWM pin
  pinMode(magPin, OUTPUT);

  // built in RGB LED
  pinMode(LEDG, OUTPUT);
  pinMode(LEDR, OUTPUT);
  pinMode(LEDB, OUTPUT);
  digitalWrite(LEDB,1);
  digitalWrite(LEDG,1);
  digitalWrite(LEDR,1);
  // Try to initialize!

  //start i2c and set speed
  Wire1.begin();
  Wire1.setClock(400000); //400khz clock
  Wire.begin();
  Wire.setClock(400000); //400khz clock

  //start up checks
  int err1 = IMU1.init(calib, IMU_ADDRESS);
  int err0 = IMU0.init(calib, IMU_ADDRESS);
  if (err1 != 0 || err0 != 0) {
    while (true) {
      blinkRedSingle();
    }
  }
  //err = IMU.setGyroRange(500);
  err1 = IMU1.setAccelRange(16);
  err0 = IMU0.setAccelRange(16);
  if (err1 != 0 || err0 != 0) {
    while (true) {
      blinkRedSingle();
    }
  }
```

```
  blinkGreen(5);
  delay(500);

  if (!SD.begin(chipSelect)) { // error if the SD card doesnt show
    while (true) {
      blinkRedSingle();
    }
  }

  blinkGreen(5); // blink the green LED to indicate that system is ready
  delay(100);
}


void loop() {
  if(!digitalRead(button2)) {
    coilMode == nModes ? coilMode = 1 : ++coilMode; // cycle up coil mode or reset if at max
    for(int i = 1; i <= coilMode; i++) { //blink the blue LED corresponding to the current
mode
      digitalWrite(LEDB, 0);
      delay(70);
      digitalWrite(LEDB, 1);
      delay(70);
    }
    delay(300);

  }
  // Start/Stop data capture
  if(!digitalRead(button1)) { //if button pressed
    if(!dataCapture) {
      dataCapture = 1; //set data capture state true
      String fileNumber = String(getLastData() + 1); // make string with new file number
      while(fileNumber.length() != 3) { // add leading zeros until 3 char long
        fileNumber = "0" + fileNumber;
      }
      String fileName = "DAT" + fileNumber + ".TXT"; //create complete file name
      dataFile = SD.open(fileName, FILE_WRITE); //open new file
      blinkGreen(10);
      dataFile.println("millis, accelX_1, accelX_0");
      digitalWrite(LEDG,0);
    } else {
      dataCapture = 0; // set data capture state false
      digitalWrite(LEDG,1); //switch to red LED until file closed and at least 1 second passed
      digitalWrite(LEDR,0);
```

```
      int timer = millis();
      dataFile.close();
      while (timer+1000 > millis()) {}; //wait a minimum of 1 second or as long as it takes to close the file
      digitalWrite(LEDR,1);
    }
  }


  // -------- UPDATE ACCELEROMETER VALUES --------
  time0 = millis();
  IMU1.update();
  IMU0.update();
  IMU1.getAccel(&accelData);
  accel1 = accelData.accelY;
  IMU0.getAccel(&accelData);
  accel0 = accelData.accelY;
  // -------- UPDATE ACCELEROMETER VALUES --------

  // calculate relative acceleration
  relAccel = accel0 - accel1; // calculate relative acceleration
  dRelAccel = 1000 * (relAccel - lastRelAccel) / (time0 - time1); // calculate rate of relative acceleration change (jerk)

  // \/ \/ \/ \/ \/ \/ \/ \/ COIL ACTUATION \/ \/ \/ \/ \/ \/ \/ \/ //
  switch(coilMode) { // set the level of coil actuation when activated
    case 1: // coil off
      analogWrite(magPin, 0);
      break;
    case 2: // coil auto on/off
      if(dRelAccel > 10 && relAccel < -(ACCEL_SEN)) { // if relative jerk positive and relative acceleration negative (accelerating and moving in compression direction)
        analogWrite(magPin, 255); // activate the coil with the 8-bit PWM value selected using the mode button
        timeOn = time0;
      } else if(dRelAccel < 0/* && relAccel > -(ACCEL_SEN)*/) { // if relative jerk negative and relative acceleration positive (accelerating and moving in rebound direction)
        if(timeOn + 200 < time0) { // if at least "X" ms have passed since the coil was last set to active
          analogWrite(magPin, 0); // deactivate coil
        }
      }
      break;
    case 3: // cycle coil in a sawtooth wave to demonstrate varied damping
      long currentTime = millis();
```

```
      if(currentTime > timeCycle + 5) {
        magPWM == 255 ? magPWM = 0 : magPWM++;
        analogWrite(magPin, magPWM);
      }
      break;

  }




  // /\ /\ /\ /\ /\ /\ /\ /\ COIL ACTUATION /\ /\ /\ /\ /\ /\ /\ /\ //


  // update previous relative acceleration and time for next iteration
  lastRelAccel = relAccel;
  time1 = time0;


  // capture a row of data if data capture active
  if(dataCapture) {
    String dataString = "";
    dataString += time0;
    dataString += ", ";
    dataString += accel1;
    dataString += ", ";
    dataString += accel0;
    dataFile.println(dataString); //add row to file
  }


} // END MAIN LOOP


//get the number from the name of the last data file written
int getLastData(void) {
  File root = SD.open("/"); //open card at root
  int lastData = 0; //create var to store highest number data file
  while (true) {
    File entry =  root.openNextFile();
    String fileName = entry.name();
    if (fileName.substring(0,3) == "DAT") { //check if is data file
      int currentN = fileName.substring(3,6).toInt();
      if(currentN > lastData) {lastData = currentN;} //update to highest data file number
      entry.close();
```

```
    }else if (! entry) {
      break;
    }
  }
  root.close();
  return lastData;
}


//blink the green led n times
void blinkGreen(int n) {
  for(int i=0; i < n; i++) {
    digitalWrite(LEDG,0);
    delay(50);
    digitalWrite(LEDG,1);
    delay(50);
  }
}


// blink the red led 1 time
void blinkRedSingle(void) {
  digitalWrite(LEDR,0);
  delay(50);
  digitalWrite(LEDR,1);
  delay(100);
}
```

A 7: Control System Code

```
function [vel, force] = protoV2_newtonian_func(ID_tube, d, d_IFP, L, d1, v_max, rho,
mu_MR)

% ------ Geometry ------
d_mean = (ID_tube + d)/2;               % mean diameter
W = pi*d_mean;
a = (ID_tube - d)/2;
A_p = pi/4 * d^2;
A_a = pi/4 * (ID_tube^2 - d^2);

A1 = pi/4 * d1^2;

% ------ Velocity range ------
vel = linspace(0, v_max, 1000);

% ------ Flow rate ------
Q = vel .* A1;
v_gap = Q ./ A_a;

% ------ Pressure gradient ------
dPdy = -(12 .* Q .* mu_MR) ./ (a^3 * W);

% ------ Pressure drop ------
deltaP_visc = abs(dPdy) .* L;

A_piston_effective = pi/4 * (ID_tube^2 - d^2);                % area of annulus
A_IFP_effective = pi/4 * (ID_tube^2 - d_IFP^2);               % area between IFP
shaft OD and cartridge tube ID

d_piston_effective = sqrt(4/pi * A_piston_effective);         % effective diameter
of annulus gap
D_IFP_effective = sqrt(4/pi * A_IFP_effective);               % effective diameter
of IFP shaft -> cartridge tube

K_con = 0.42 * (1 - d_piston_effective^2/ID_tube^2);          % sudden contraction
loss coefficient
K_exp = (1 - d_piston_effective^2/D_IFP_effective^2);         % sudden expansion
loss coefficient

deltaP_loss = 0.5 * (K_con + K_exp) * rho * v_gap.^2;         % pressure drop due
to minor losses
deltaP = deltaP_visc + deltaP_loss;                           % total pressure drop
= (viscous ΔP) + (minor loss ΔP)

% ------ Force ------
force = deltaP .* A_p;

end
```

```
%protoV2_newtonian_func_call

clear; clc;

% ------ Constants ------
ID_tube = 20.2e-3;
L = 20e-3;
d1 = 10e-3;
v_max = 8;
rho = 2650;
mu_MR = 0.055;
d_IFP = 8e-3;

% ------ Vector of piston diameters to test ------
d_vec = [17.5e-3, 18.0e-3, 18.5e-3, 19.0e-3, 19.5e-3];
a_vec = (pi / 4) .* d_vec .^2;

% ------ Velocities to report ------
v_report = 1:8;                         % 1–8 m/s
F_table = zeros(length(d_vec), length(v_report));

figure;
hold on;

legend_strings = strings(length(d_vec),1);

for i = 1:length(d_vec)

    d = d_vec(i);

    [vel, force] = protoV2_newtonian_func(ID_tube, d, d_IFP, L, d1, v_max, rho,
mu_MR);

    plot(vel, force, 'LineWidth', 2);

    legend_strings(i) = sprintf('d_p = %.2f mm', d*1000);

    % ------ Interpolate force at desired velocities ------
    F_table(i,:) = interp1(vel, force, v_report);

end

legend(legend_strings, 'Location', 'northwest');

grid on;
xlabel('Axel Velocity (Fluid vel. = 0.245*(Axel vel.)) [m/s]');
ylabel('Force [N]');
title('Plane Poiseuille Flow Model Approx.');
box on
hold off;

d_mm = d_vec' * 1000;   % Convert piston diameters to mm for display
a_vec;
a_mm_2 = a_vec * 1e6;
```

```
% ------ Create table -------
Force_Table = array2table(F_table, ...
    'VariableNames', compose('v_%d_[m/s]', v_report));

Force_Table = addvars(Force_Table, d_mm, ...
    'Before', 1, ...
    'NewVariableNames', 'Piston_Diameter_mm');

disp(Force_Table);
```

```
function NonlinearMagneticV6(plot)


%input 1 for heat map of magnetic flux, input 2 for vector lines

%%BH curve incorporated for fluid
%%Steel components are linear, relative permeability of 1000
%%Trim area for poster plot

%Define geometry of steel and aluminum components, copper windings, and
%fluid space

R1=[3,4,0,.020,.020,0,0,0,.030,.030]';
R2=[3,4,0,.0095,.0095,0,.005,.005,.010,.010]';
R3=[3,4,0,.005,.005,0,.010,.010,.020,.020]';
R4=[3,4,0,.0095,.0095,0,.020,.020,.025,.025]';
R5=[3,4,0,.0025,.0025,0,.025,.025,.030,.030]';
R6=[3,4,.005,.0095,.0095,.005,.010,.010,.020,.020]';
R7=[3,4,.0101,.011,.011,.0101,0,0,.030,.030]';
R8=[3,4,.011,.015,.015,.011,.005,.005,.025,.025]';
ns=char('R1','R2','R3','R4','R5','R6','R7','R8');
sf='R1+R2+R3+R4+R5+R6+R7+R8';
gdm=[R1, R2, R3, R4, R5, R6, R7, R8];
g=decsg(gdm,sf,ns');
%pdegplot(g,FaceLabels="on")

%Define model and relative permeabilities of materials
model=femodel(AnalysisType="magnetostatic",Geometry=g);
model.PlanarType = "axisymmetric”;
model.VacuumPermeability = 1.2566370614E-6;
model.MaterialProperties = materialProperties(RelativePermeability=1);
model.MaterialProperties(1)=materialProperties(RelativePermeability=400);
model.MaterialProperties(3)=materialProperties(RelativePermeability=1000);
model.MaterialProperties(4)=materialProperties(RelativePermeability=1000);
model.MaterialProperties(5)=materialProperties(RelativePermeability=1000);
model.MaterialProperties(9)=materialProperties(RelativePermeability=1000);

%Calculate current density based of phyiscal dimensions, number of turns,
%wire guage, and current flow
N=160;
I=2;
conductorDiameter = .0159 * .0254; %convert to m
conductorArea = (conductorDiameter/2)^2*3.14159;
wireDiameter = .017 * .0254; %convert to m
wireArea = (wireDiameter/2)^2*3.14159;
totalArea = wireArea * N;

density = I/conductorArea;
```

```
areaFactor = totalArea/(.004*.010);

%Define current density of copper winding face
%Define boundary conditions
model.FaceLoad(7)=faceLoad(CurrentDensity=density*areaFactor);
%pdegplot(model.Geometry,EdgeLabels="on")
model.EdgeBC(1)=edgeBC(MagneticPotential=0);
model.EdgeBC(5)=edgeBC(MagneticPotential=0);
model.EdgeBC(6)=edgeBC(MagneticPotential=0);
model.EdgeBC(7)=edgeBC(MagneticPotential=0);
model.EdgeBC(8)=edgeBC(MagneticPotential=0);
model.EdgeBC(9)=edgeBC(MagneticPotential=0);
model.EdgeBC(18)=edgeBC(MagneticPotential=0);
model.EdgeBC(19)=edgeBC(MagneticPotential=0);
model.EdgeBC(20)=edgeBC(MagneticPotential=0);
model.EdgeBC(21)=edgeBC(MagneticPotential=0);
model.EdgeBC(22)=edgeBC(MagneticPotential=0);
model.EdgeBC(23)=edgeBC(MagneticPotential=0);
model.EdgeBC(24)=edgeBC(MagneticPotential=0);

%create finite element points
internalEdges = faceEdges(model.Geometry,2:8);
model=generateMesh(model,Hedge={internalEdges,.0007});
%pdemesh(model)

%Solve model with MR fluid characterized as a linear medium
Rlin=solve(model);
mu0=1.2566370614E-6;

%Define BH curve of MR fluid and interpolate to a function
B=[0 0.05 0.1 0.15 0.2 0.25 0.3 0.35 0.4 0.45 0.5 0.55 0.6 0.65 0.7];
H=[0 8966 16552 24828 34483 42759 52414 62759 74483 86897 102069 122759 152414 185517
226897];

HofB= griddedInterpolant(B,H,'makima','linear');
muR = @(B) B./HofB(B)/mu0;


%Reevaluate model with non-linear characteristics of MR fluid, and linear
%solution as a starting condition for system
model.MaterialProperties(1)=materialProperties(RelativePermeability=@(~,s)
muR(s.NormFluxDensity));

model.FaceIC=faceIC(MagneticVectorPotential=Rlin);

R=solve(model);

%Plot data based of function input
```

```
Bmag = sqrt(R.MagneticFluxDensity.Bx.^2 + R.MagneticFluxDensity.By.^2);

if plot == 1
    pdeplot(R.Mesh,XYData=Bmag)
else
    x=0:0.0008:.020;
    y=0:0.0008:.030;
    [X,Y]=meshgrid(x,y);
    intrpBcore = R.interpolateMagneticFlux(X,Y);

    Bx=reshape(intrpBcore.Bx,size(X));
    By=reshape(intrpBcore.By,size(Y));

    quiver(X,Y,Bx,By,Color="r")
end

hold on
pdegplot(model.Geometry);


%Calculate average magnetic flux strength within active region
x1=.0095:.0001:.0101
y1=.020:.0002:.025;
[X1,Y1]=meshgrid(x1,y1);
intrpBcore = R.interpolateMagneticFlux(X1,Y1);

Bx1=reshape(intrpBcore.Bx,size(X1));
By1=reshape(intrpBcore.By,size(Y1));

B=zeros(26,7);
Bavg = 0;
for i=1:26
    for j=1:7
        B(i,j)=sqrt(Bx1(i,j)*Bx1(i,j)+By1(i,j)*By1(i,j));
        Bavg = Bavg + B(i,j);
    end
end

Bavg = Bavg / (26*7)


%% Wire Diameter 0.017 in
%% Conductor Diameter 0.0159 in

%B 1A 0.1096
%B 2A 0.2194
%B 4A 0.4390
```